

HACKXPLORE 2026

Challenge Overview -- Rated for a dev/R&D jury

June 26-28, Karlsruhe | 36h | ~150 participants | Jury: mostly R&D / engineering

JUDGING RUBRIC

25% Creativity | 25% Realisability | 25% User-centricity & Usability | 25% Pitching

Note: R&D/dev jury weights technical credibility and realisability higher than a business jury would. Working demos and clean architecture matter. Flashy slides without substance get called out in Q&A.

1. ZEISS Innovation Hub @ KIT -- EEG Mind Control *** ONLY 3 SPOTS ***

Brain signals replace keyboard strokes -- control Canabalt / 2048 with your mind

Category: Signal Processing + ML

ZEISS provides EEG headset. Process live brain data to detect intended keystrokes and classify them. Noisy time-series classification problem. They ran this challenge in 2025 -- they know what a working demo looks like and will spot a fake.

EEG hardware provided, scipy/sklearn signal chain, real-time classification



Why: Only pick if you have signal processing experience. Highest ceiling, lowest floor.

2. LBBW -- The Future of Giving (~4 spots)

Digital Impact Account that grows from micro-giving to your own foundation

Category: Product/UX + FinTech

Build a clickable prototype for a Donor-Advised Fund product aimed at 20-45 year olds. NOT a donation app -- a banking product that makes giving identity-driven. Two user groups: growing-wealth millennials + post-exit founders. They explicitly say "answer it in the product, not on slides."

Figma/prototype, user journey design, business model sketch



Why: Strongest on UX scoring. Weakest on technical depth -- dev jury will probe the business logic.

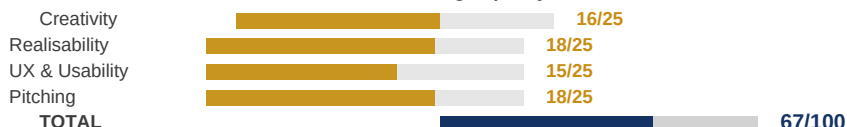
3. Hitachi Rail -- From Paper to Data (~4 spots)

OCR pipeline for heterogeneous railway planning documents

Category: OCR + Data Modeling + Standardization

PT1 planning docs arrive as stamped/scanned paper in wildly varying formats. Extract tables, text, and topological relationships. Transform into standardized format for a central database. High data quality mandatory. Intelligent detection, not brute-force OCR.

Document AI, table extraction, data schema design, quality validation



Why: Solid but unremarkable. Dev jury will appreciate the engineering but it will not stand out.

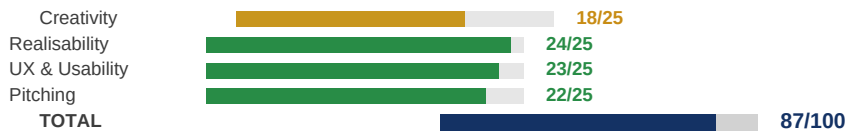
4. VEGA Grieshaber KG -- Smart Watering for Urban Trees (~4 spots)

ESP32 soil sensors + citizen engagement platform + municipal dashboard

Category: IoT + Sustainability + Smart City

City of Karlsruhe invests heavily in tree irrigation but no one knows which trees actually need water. Build a platform: soil moisture sensors (ESP32 + capacitive sensor provided) feed a dashboard. Citizens get push notifications. Municipalities get data-driven decisions. The hardware is trivial -- the platform is where you win.

ESP32 firmware (basic), data pipeline, dashboard UI, citizen notification system



Why: Highest realisability score. Working project > working prototype. Dev jury wants to touch it.

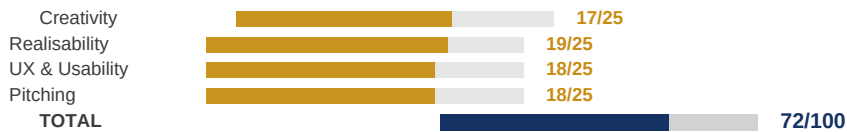
5. Rentschler Biopharma -- Digital Workflow Transformation (~4 spots)

Handwritten pharma batch records -- extract, structure, output to Excel

Category: OCR + Local AI + Compliance

Each pharma batch generates 20 forms (100-200 pages) of handwritten records. Extract parameters and values from varying form layouts. CRITICAL: must run LOCALLY -- no cloud. Output to Excel. Bonus: anomaly detection, human review interface.

Handwriting OCR, local-only deployment, form layout variance, Excel output



Why: Solid engineering challenge. The local-only constraint is both a hard requirement and a story.

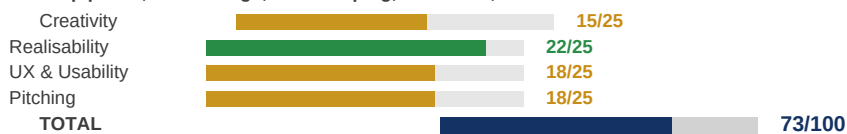
6. Hirth Engines -- AI-Powered Two-Stroke Knowledge Database (~4 spots)

RAG + semantic search + chatbot for fragmented engine knowledge

Category: RAG + Knowledge Management + AI

Two-stroke engine knowledge is scattered across forums, manuals, and personal experience. Build an AI system that collects, structures, and intelligently serves this knowledge. Semantic search, chatbot, recommendation system. Small company (~12 people) -- your solution will actually be looked at.

RAG pipeline, embeddings, web scraping, vector DB, chat interface



Why: Safest technical bet. But 3 teams will build the same thing. Differentiation is the challenge.

RANKINGS -- BY TOTAL SCORE (DEV/R&D JURY)

1st	VEGA	87/100	Everything works. Real hardware, real impact, 3 user groups. Dev jury will want to touch the sensors.
2nd	LBBW	78/100	Best UX score. Weakest technical depth -- prepare for business-logic Q&A.
3rd	Hirth	73/100	Safe, well-understood. Problem: 3 teams will build the exact same thing.
4th	ZEISS	72/100	Highest ceiling, lowest floor. 3 spots. Hardware cooperation decides your fate.
5th	Rentschler	72/100	Solid engineering. Handwriting OCR is genuinely hard -- accuracy wins.
6th	Hitachi	67/100	Backend-heavy, limited UX surface. Good engineering, hard to make exciting.

Scoring adjusts for R&D/dev jury: technical credibility and realisability carry more weight than they would with a business jury. Scores assume the team has general AI/software engineering skills and no specialized hardware experience (ESP32 is learnable; EEG signal processing is not).

Jury composition inferred from partner company reps + NXTGN/MESH board. All industry practitioners, no academics. They value working demos.